Use the following table to answer the questions.

X	3	14	48	53	67	82
P(X≤x)	.08	.15	.41	.70	.95	1

a) Find the probability that X is at most 50.
P(X≤50) = P(X≤48) = 0.41

b) Find the probability that X is between (inclusive) 20 and 67.
P(20≤X≤67) = P(X≤67) – P(X≤14) = 0.95 – 0.15 = 0.80

c) Construct the corresponding pdf.

X	3	14	48	53	67	82
P(X=x)	0.08	0.07	0.26	0.29	0.25	0.05

d) Calculate the mean of X.

E(X) = 3*.08 + 14*.07+ 48*.26 + 53*.29 + 67*.25 + 82*.05 = 49.92

e) Calculate the standard deviation of X.

$Var(X) = 3^2*.08 + 14^2*.07+ 48^2*.26 + 53^2*.29 + 67^2*.25 + 82^2*.05 - 49.22^2$

Var(X) = 2886.54-2492.0064=394.5336

$\sigma = \sqrt{394.5336} = 19.8628$